AI- Based Presentation Skills Evaluation System

Supervisor: Dr. Fatma Helmy
Supervising Assistant: Eng.Saja Saadoun
Team Members: Ahmed Yehia, Arwaa Mamdoh, Mayar Adel, Mostafa Wael

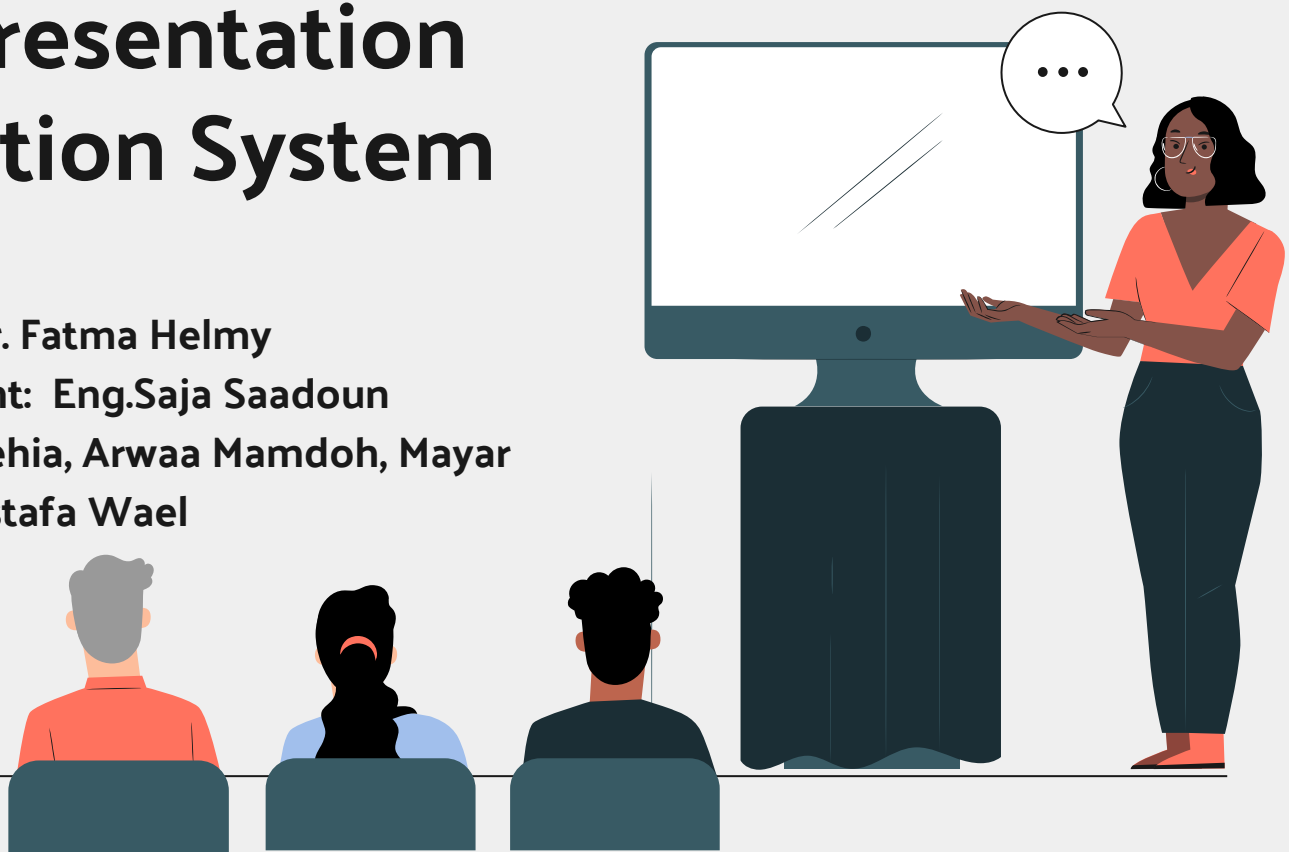


Table of contents

01 INTRODUCTION

02 MOTIVATION

03 PROBLEM STATEMENT

04 SYSTEM OVERVIEW

05 SIMILAR SYSTEMS

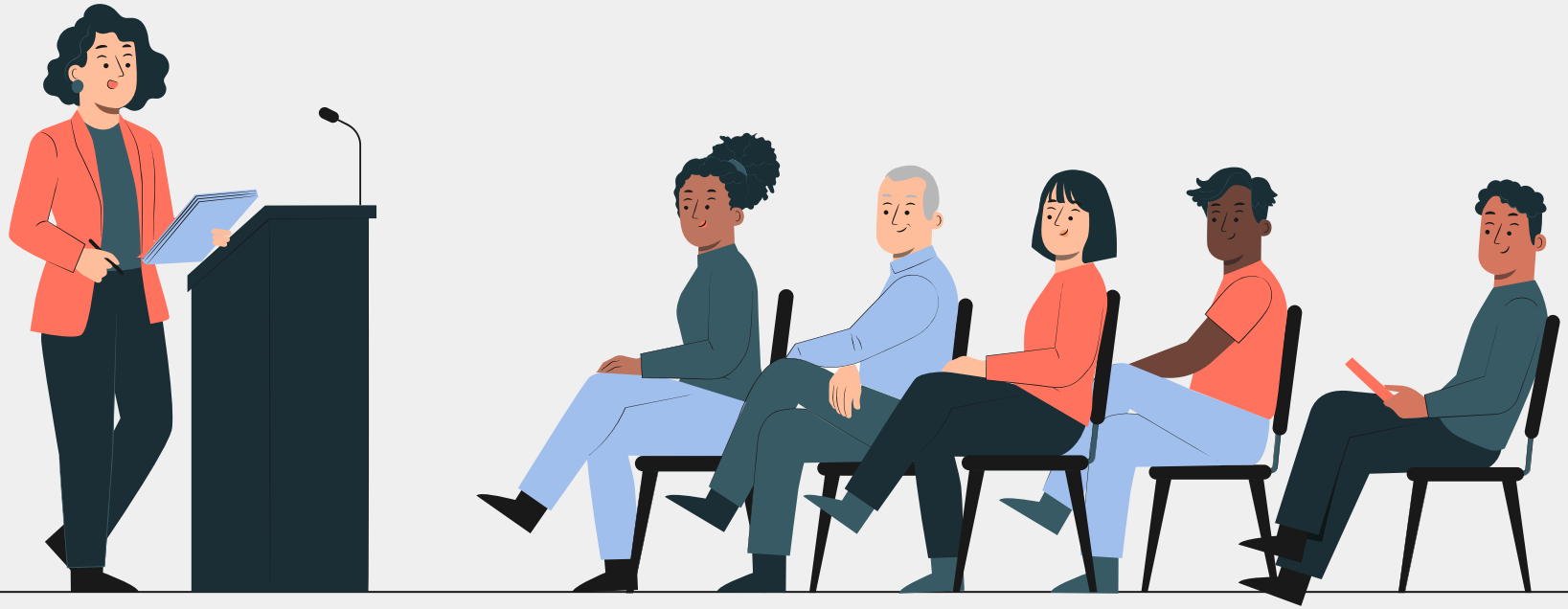
06 DELIVERABLES

07 TIMELINE

08 SUPPORTIVE DOCUMENTS

09 DEMO

01 INTRODUCTION



01 INTRODUCTION

- Presentation skills are essential for success in academic and professional settings
- People often lack access to feedback on body language, voice tone, and clarity.
- The AI-powered app analyzes presentation videos and provides personalized feedback on non-verbal cues, facial expressions, and speaking style
- It helps users improve their confidence and communication skills.

02 MOTIVATION

02 MOTIVATION

❖ ACADEMIC

- The system helps students and academics enhance their communication skills.
- It provides accurate and measurable insights into their presentation style.
- These insights enable users to refine their ability to engage with audiences effectively.
- It supports better idea sharing and fosters improved communication overall.
- So enhancing presentation skills boosts users' confidence.

02 MOTIVATION

❖ BUSINESS

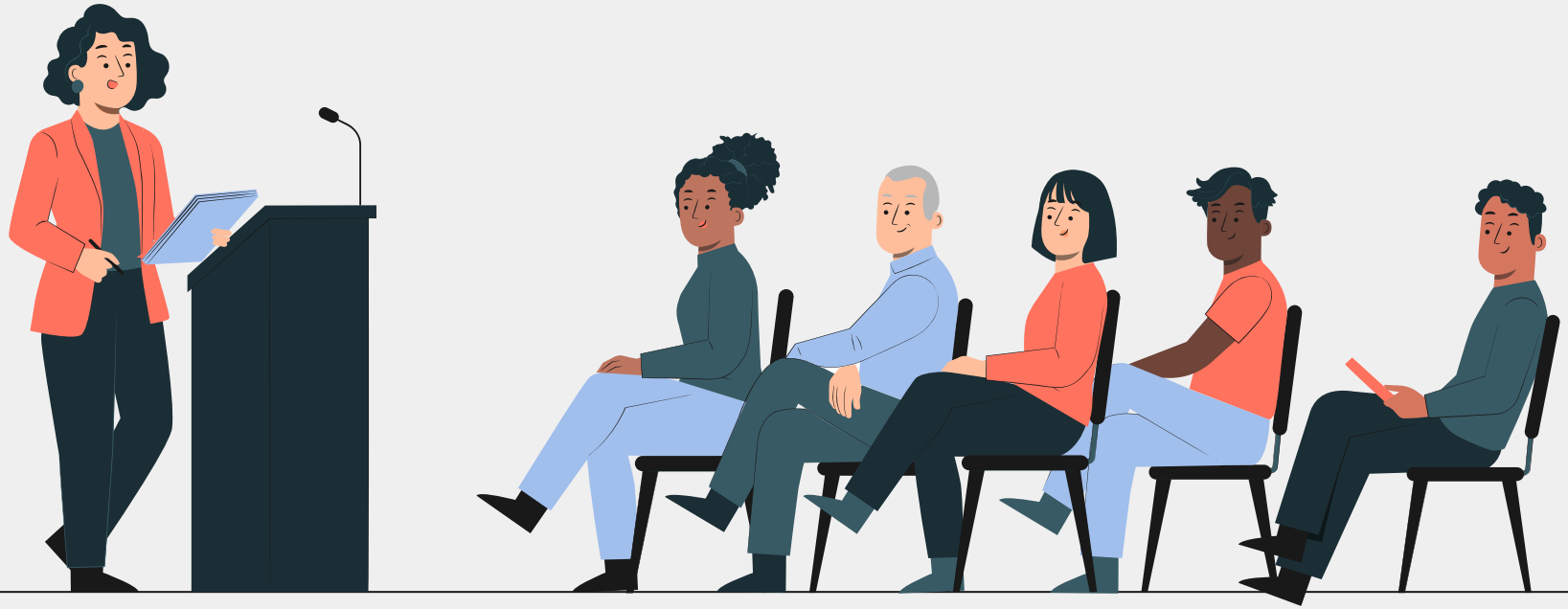
- Effective communication is essential for business success.
- It plays a crucial role in sales meetings, client presentations, and leadership discussions.
- Delivering a compelling message can significantly impact outcomes.
- The system assists professionals in assessing and enhancing their communication skills.
- It aims to improve users' confidence in professional interactions.

03 PROBLEM STATEMENT

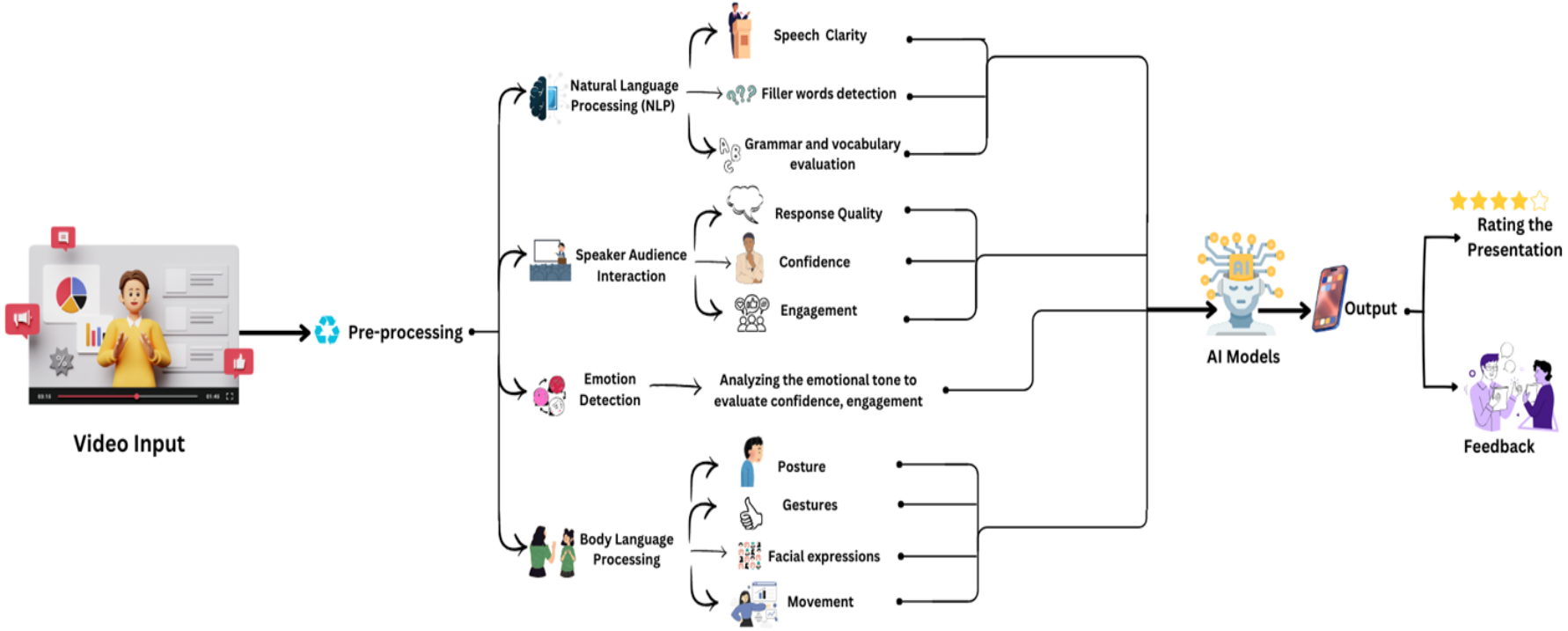
03 PROBLEM STATEMENT

- Speakers have difficulty recognizing and addressing areas of weakness, such as body language and speech clarity.
- Many people cannot afford or obtain professional coaching.
- Tracking progress over time can be difficult.
- There is a need for personalized, accessible tools to address these gaps.
- Evaluation of audience interaction and Q&A responses is absent from current solutions.

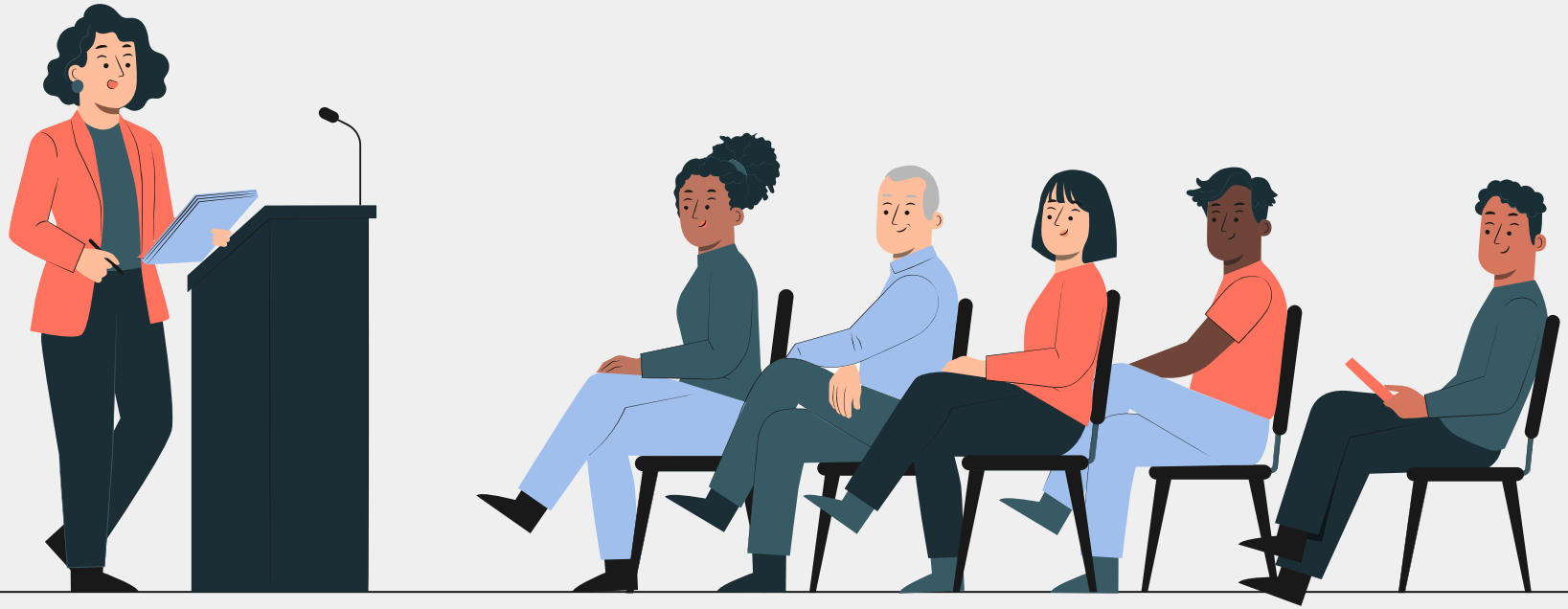
04 SYSTEM OVERVIEW



04 SYSTEM OVERVIEW



05 SIMILAR SYSTEMS

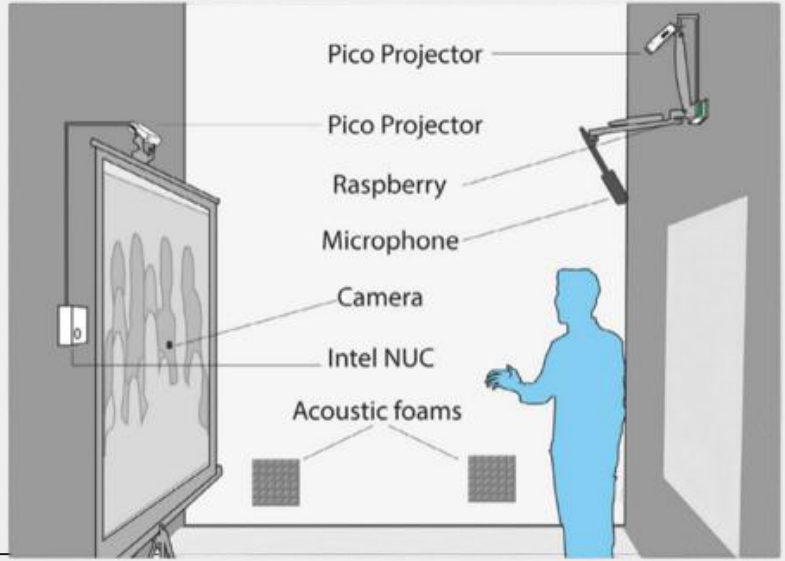


05 SIMILAR SYSTEMS

❖ Academic

Controlled evaluation of a multimodal system to improve oral presentation skills in a real learning setting

- **RAP System Overview**: Assesses oral presentation skills using multimodal learning analytics (MmLA).
- **Accuracy**: 85% accuracy in performance evaluation.
- **Student Feedback Recall**: Increased from 81% to 84% in follow-ups.
- **Self-assessment Decline**: Dropped from 74% to 68% over time.
- **Algorithms**: Employ SVM or RNN

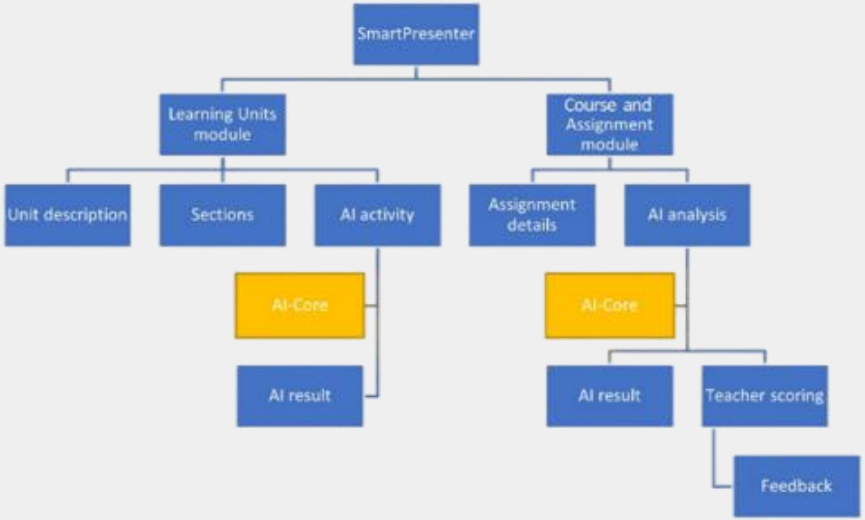


05 SIMILAR SYSTEMS

❖ Academic

AI-Assisted Enhancement of Student Presentation Skills:
Challenges and Opportunities

- **AI-Assisted Platform**: Provides real-time feedback on presentation skills.
- **Speech Analysis**: Assesses fluency, pronunciation, facial expressions, and vocal fillers.
- **Correlation Scores**: Fluency (0.136), pronunciation (0.450).
- **Limitations**: AI struggles with fluency and emotion recognition compared to human evaluations.
- **Algorithms**: Uses speech-to-text, acoustic features, and facial recognition (CNN) for real-time feedback.

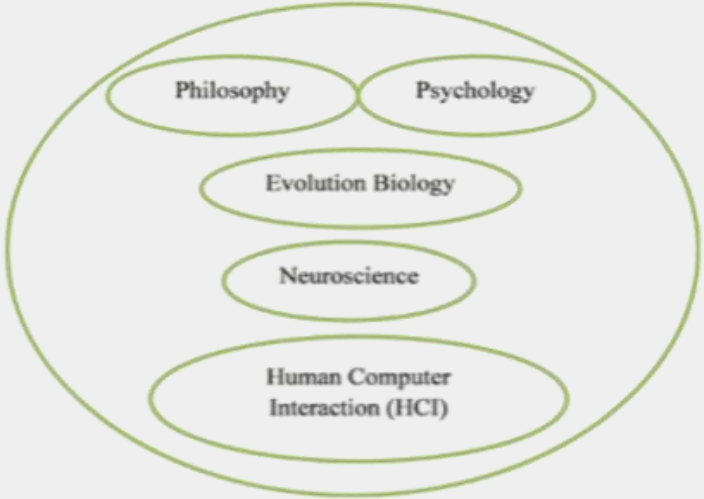


05 SIMILAR SYSTEMS

❖ Academic

Machine learning for human emotion recognition: a comprehensive review

- **Naive Bayes (NB)**: Accuracy 74%.
- **SVM**: Average accuracy 85.47%.
- **Random Forest (RF)**: Accuracy 95.6%.
- **CNN**: Accuracy 80%
- **Decision Tree (DT)**: Accuracy 74%.
- **Bi-GRU**: Accuracy 79.46%.
- **Ensemble Learning**: Stacking achieved 98.2% accuracy.



05 SIMILAR SYSTEMS

❖ Business Applications

Orai

AI Speech Coach Assistant

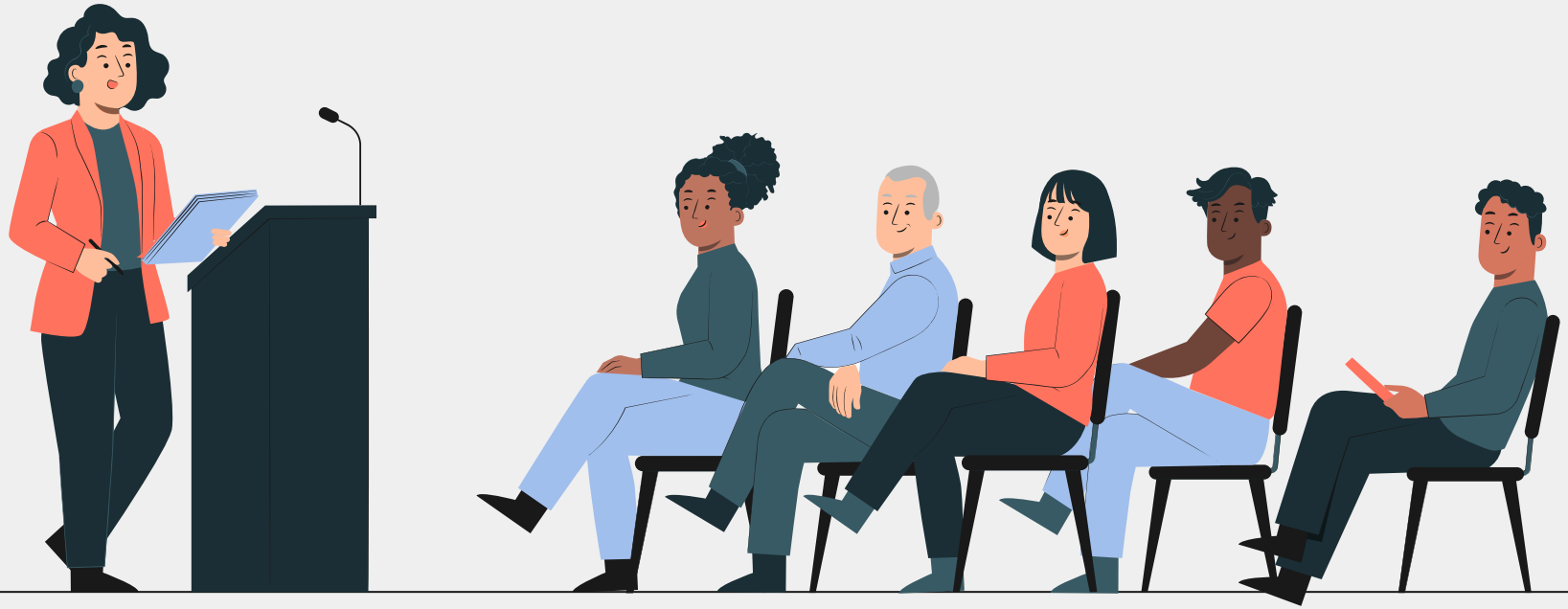
Speeko

Voice Recognition & Feedback

Virtual Speech

VR Speech Training Platform

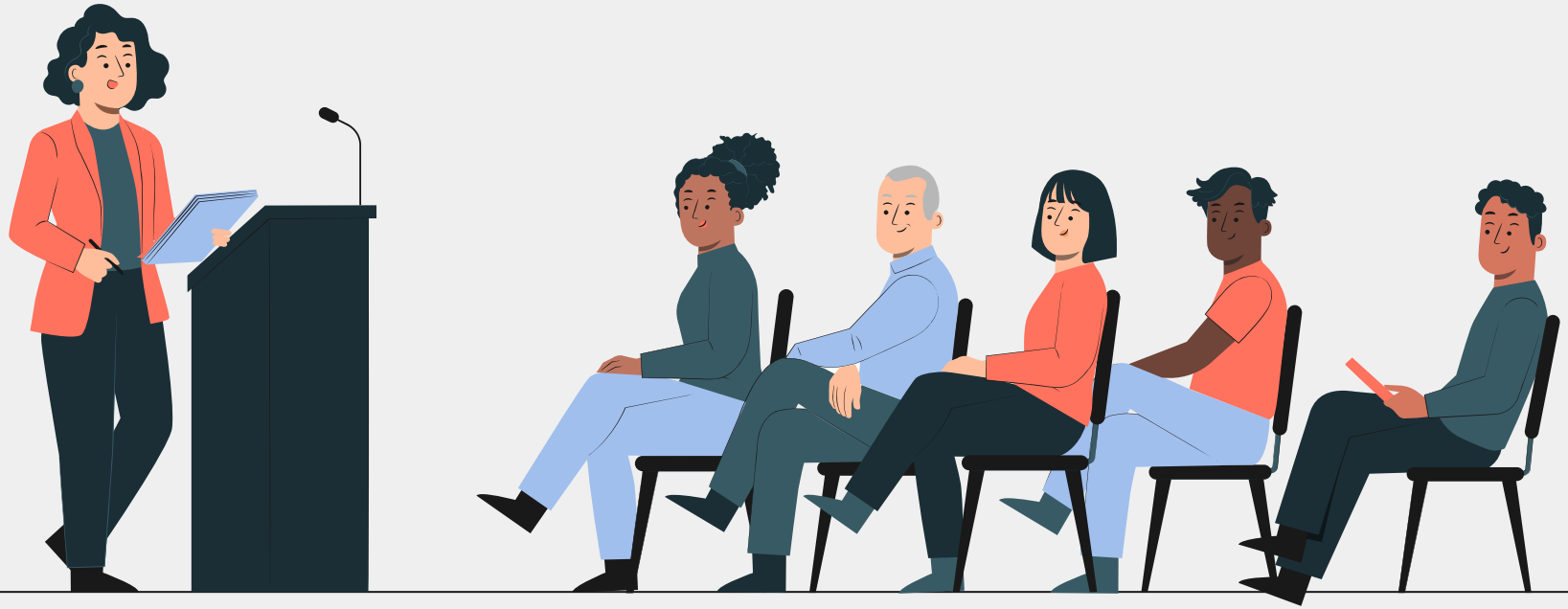
06 DELIVERABLES



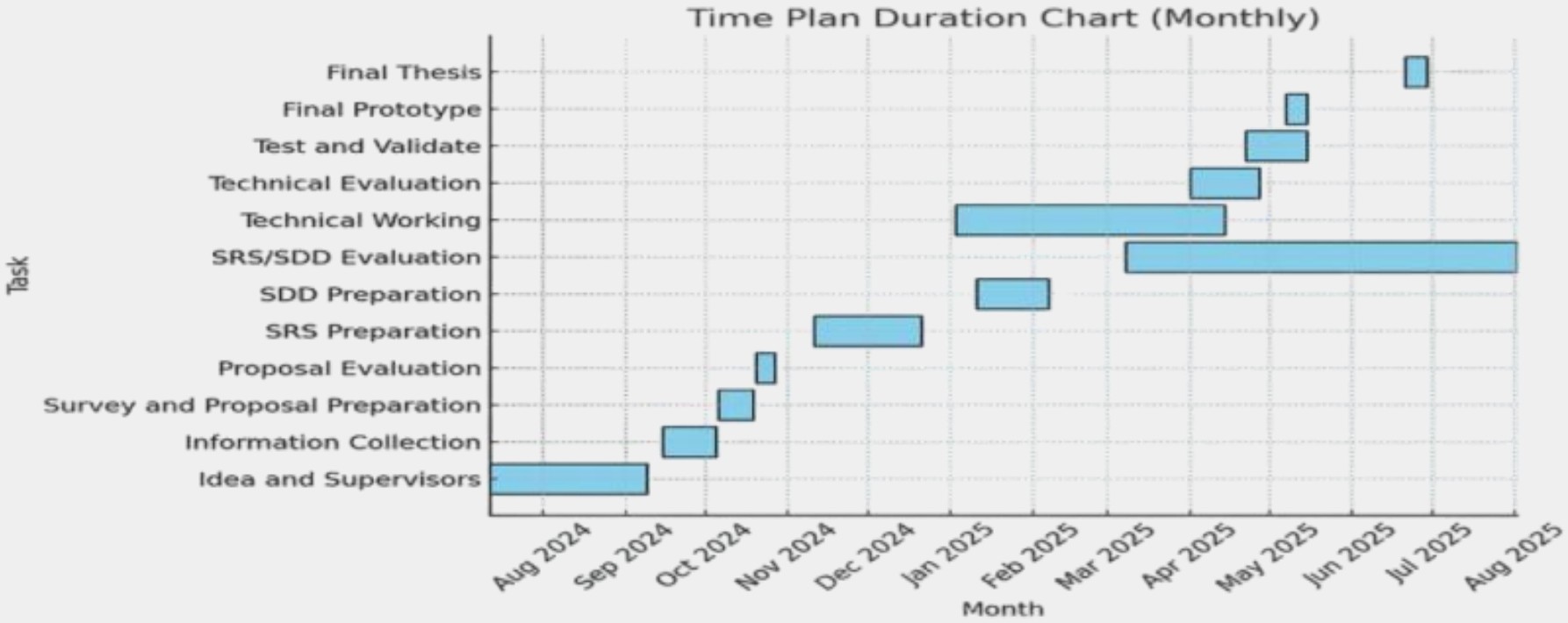
06 DELIVERABLES

- Proposal document
- Academic Papers
- SRS Document
- SDD Document
- Final Thesis Document
- Mobile Application

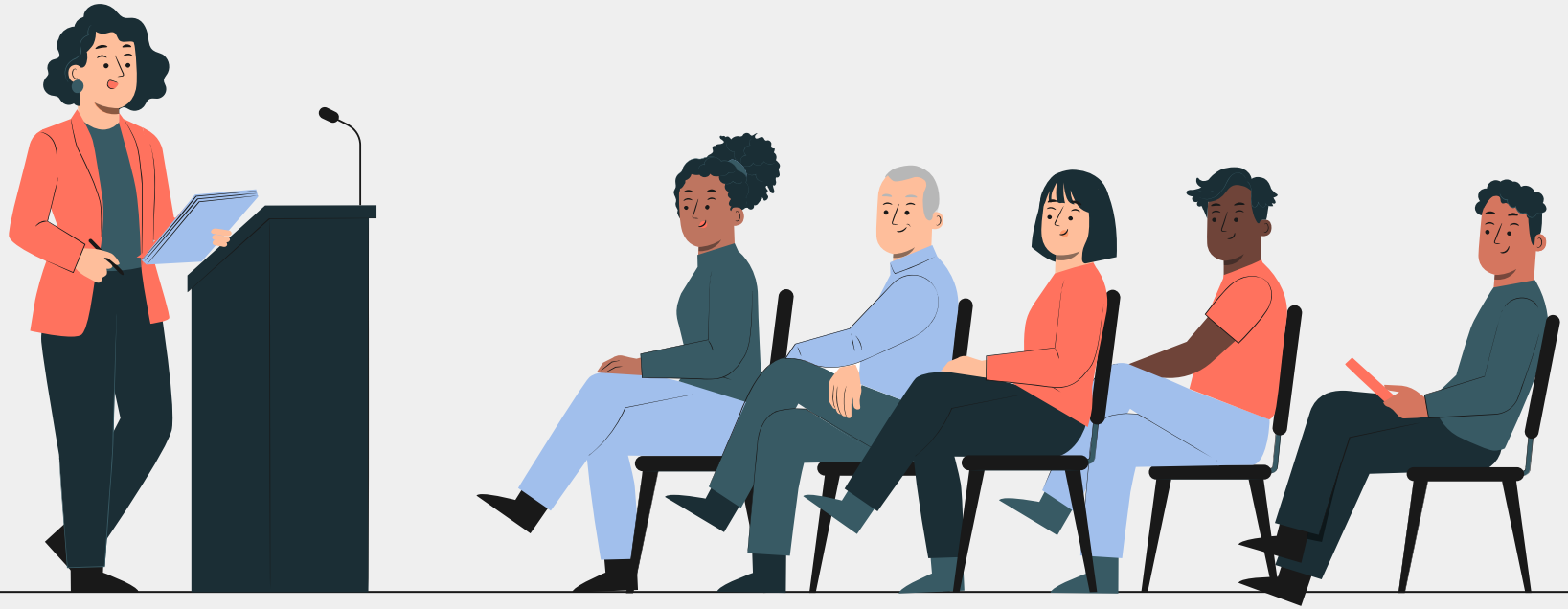
07 TIME PLAN



07 TIME PLAN


Time Plan Duration Chart (Monthly)
Task
Final Thesis
Final Prototype
Test and Validate
Technical Evaluation
Technical Working
SRS/SDD Evaluation
SDD Preparation
SRS Preparation
Proposal Evaluation
Survey and Proposal Preparation
Information Collection
Idea and Supervisors
Aug 2024
Sep 2024
Oct 2024
Nov 2024
Dec 2024
Jan 2025
Feb 2025
Mar 2025
Apr 2025
May 2025
Jun 2025
Jul 2025
Aug 2025
Month

08 SUPPORTIVE DOCUMENTS



08 SUPPORTIVE DOCUMENTS

❖ Audio Dataset

- The dataset contains 5 folders, each representing different emotion:(Angry ,Fear,Happy,Sad,Neutral)
- The dataset is balanced, each folder contains 1669 .wav audio files.
- Audios length from 2 to 5 seconds, representing real-life emotional expressions.
- Designed for speech emotion recognition
- Includes labeled data for accurate emotion detection.

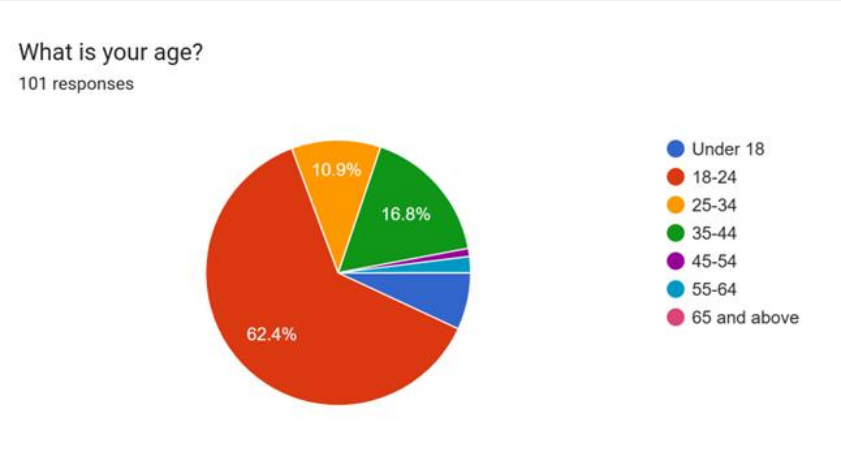
08 SUPPORTIVE DOCUMENTS

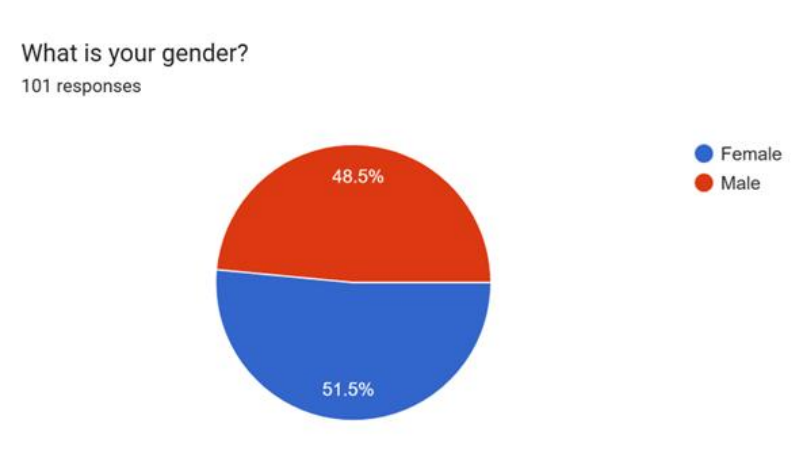
❖ Facial Expressions Dataset

- The dataset is large and well-organized, containing numerous facial screenshots from diverse individuals.
- Images are 48x48 grayscale, with faces mostly centered.
- It is divided into train and test folders, each containing 7 emotion-specific folders (Anger, Disgust, Fear, Happy, Sad, Surprise, Neutral).
- The training set includes 28,709 examples, and the public test set has 3,589 examples.

08 SUPPORTIVE DOCUMENTS

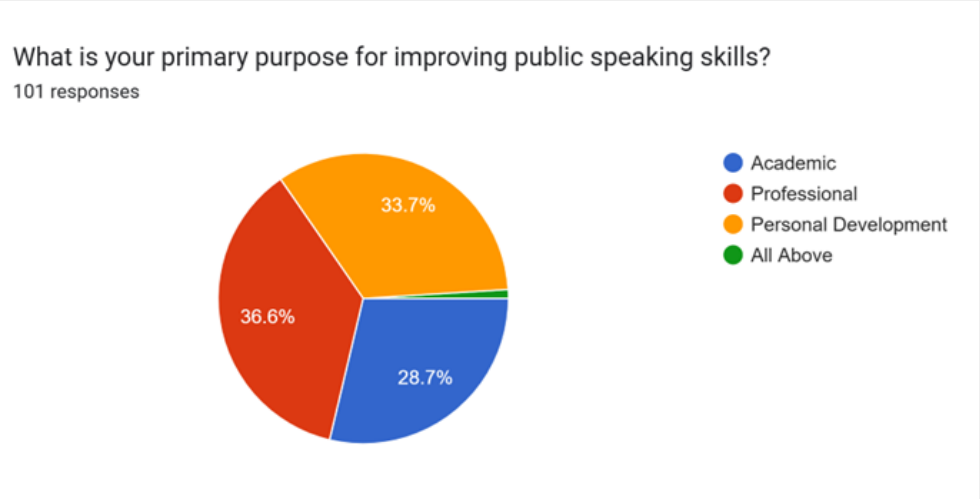
❖ Survey

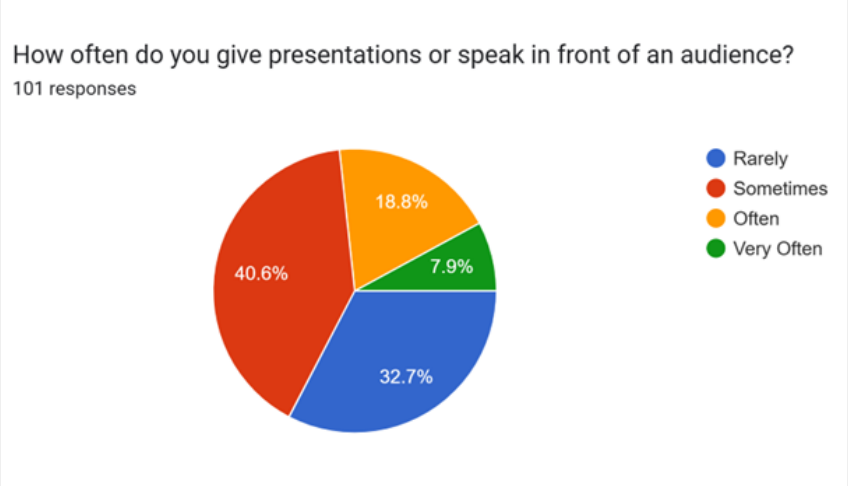




08 SUPPORTIVE DOCUMENTS

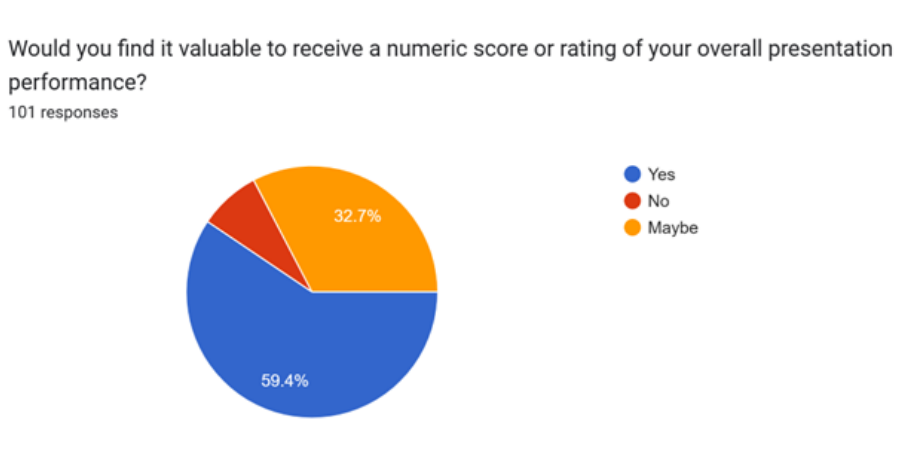
❖ Survey

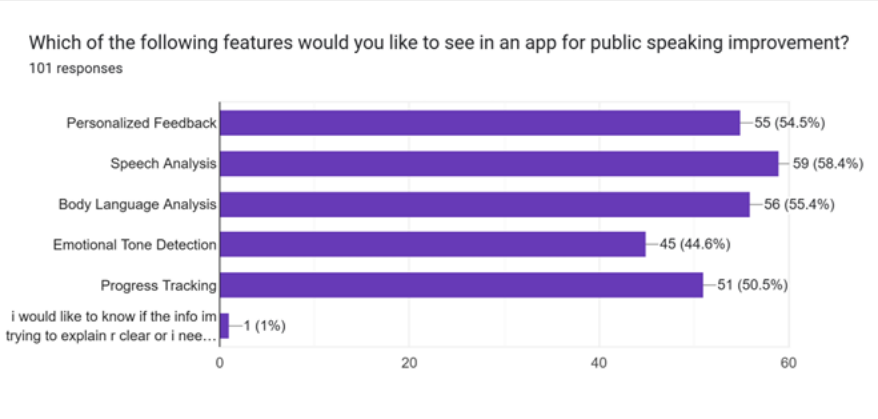




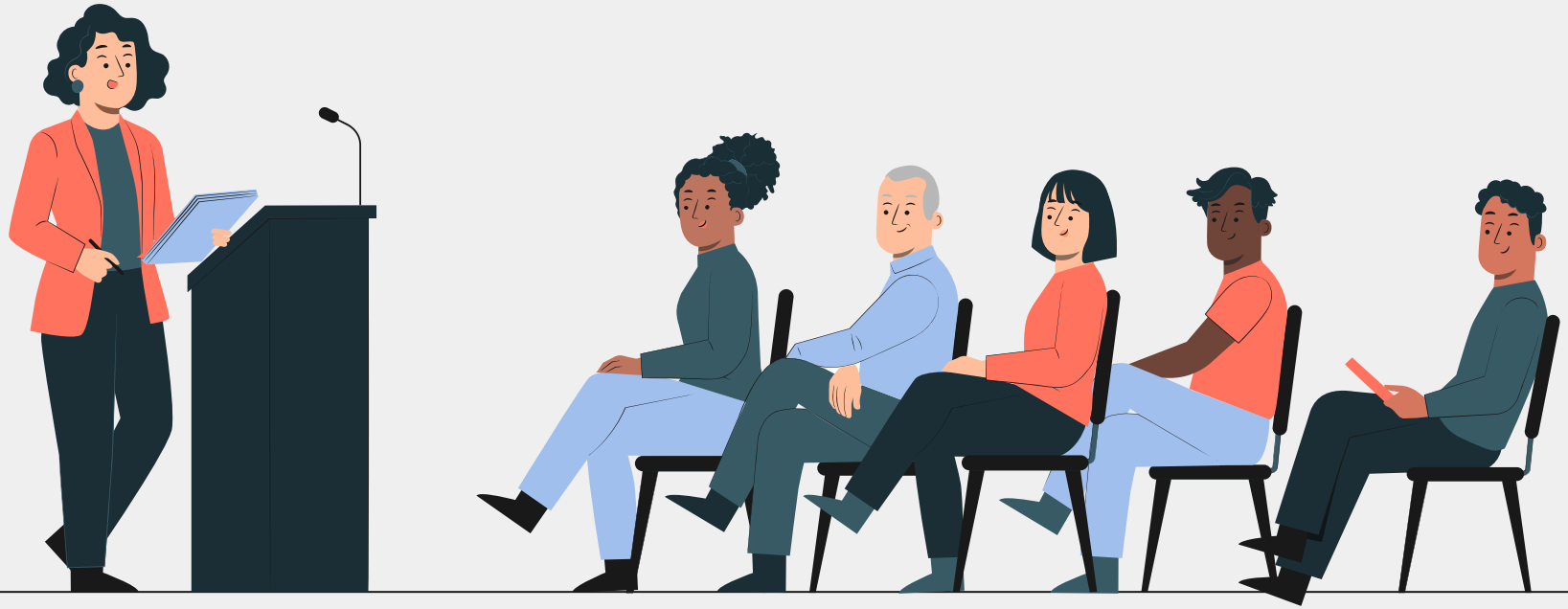
08 SUPPORTIVE DOCUMENTS

❖ Survey





09 DEMO



grammar.py U

grammar.py > ...

```python
import speech_recognition as sr
from pydub import AudioSegment
from transformers import T5ForConditionalGeneration, T5Tokenizer
import difflib
import numpy as np
import spacy
import string
import aifc

# Initialize recognizer class (for recognizing speech)
recognizer = sr.Recognizer()

# Path to your audio file (change the path as needed)
# audio_file_path = "D:/4th Year 1st Term/Graduation Project/Presentation-Skills/WhatsApp Audio 2024-10-21 at 8.55.53 PM.wav"
audio_file_path = "D:/4th Year 1st Term/Graduation Project/Presentation-Skills/presentation-test.wav"

# Load the audio file using pydub to get duration
audio_segment = AudioSegment.from_file(audio_file_path)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS COMMENTS

Python: grammar

```
Transcription: lifelong learning is a commitment to growth that transcends age occupation and personal Circumstance in the world that changes so quickly the ability to adapt
and acquire new skills is essential lifelong learning isn't just about formal education it's about maintaining curiosity embracing challenges and seeking knowledge in everyda
y life this is mindset empowers us to stay engaged relevant and resilient in the face of change whether it's learning a new language mastering craft or simply reading lifelon
g learning builds confidence expands or perspectives and connects to a broader world it encourages innovation helping us to solve complex problems and develop creative Soluti
ons most importantly a train enriches our lives and our Communities making a small compassionate and informed so let's Embrace the journey of learning for it's not just about
 what we gain but who will become along the
Your grammatical score was: 8/10
Speech Pace (WPM): 138.44480337541614
Feedback: Your pace is perfect.
PS D:\4th Year 1st Term\Graduation Project\Presentation-Skills>
```

Thanks!

Do you have any questions?